

Protein Folding – Number of Qubits

Ruihao Li

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Based on the coarse-grained model discussed in [A. Roberts et al., npj Quantum Inf. 7 \(2021\)](#), the total number of qubits needed to represent an N -amino acid peptide is the sum of the configuration qubits and the interaction qubits.

The number of configuration qubits is given by $2(N - 3) - 1$ (considering only the main chain) for sparser encoding (see discussion in the Supplementary Information of the paper). However, the authors did not give an explicit expression for the number of interaction qubits. The key to derive it is through Eq. (29) in SI:

$$H^{(1)} = \sum_{i=1}^{N-4} \sum_{\substack{j \geq i+5, \\ j-i \equiv 1 \pmod{2}}}^N h_{ij}^{(1)}.$$

For example, we consider $N = 6$. In this case, $i \in \{1, 2\}$. When $i = 1$, the only possible j is 6. When $i = 2$, no value of j is allowed. Therefore, the number of configuration qubits is 1. Next, when $N = 7$, $i \in \{1, 2, 3\}$. When $i = 1$, possible value(s) of j is 6; when $i = 2$, possible value(s) of j is 7; when $i = 3$, no j is allowed. So the number of configuration qubits is 2. Following along this line, we can find the following examples:

N	$n = N - 5$	# interaction qubits (a_n)
6	1	$1 = n^2$
7	2	$2 = n(n - 1)$
8	3	$4 = (n - 1)^2$
9	4	$6 = (n - 1)(n - 2)$
10	5	$9 = (n - 2)^2$
11	6	$12 = (n - 2)(n - 3)$
12	7	$16 = (n - 3)^2$
13	8	$20 = (n - 3)(n - 4)$
14	9	$25 = (n - 4)^2$
...	...	...

We may write the general form of a_n as $a_n = (n - a)(n - b)$, where a and b satisfy

$$\begin{aligned} a + b &= n - 1, \\ b - a &= \text{mod}(n - 1, 2). \end{aligned}$$

Solving for a and b , we get

$$\begin{aligned} a &= \frac{n - 1 - \text{mod}(n - 1, 2)}{2}, \\ b &= \frac{n - 1 + \text{mod}(n - 1, 2)}{2}. \end{aligned}$$

Therefore, we have

$$\begin{aligned}
a_n &= \left(n - \frac{n-1 - \text{mod}(n-1, 2)}{2} \right) \left(n - \frac{n-1 + \text{mod}(n-1, 2)}{2} \right) \\
&= \frac{n+1 + \text{mod}(n-1, 2)}{2} \cdot \frac{n+1 - \text{mod}(n-1, 2)}{2} \\
&= \frac{(n+1)^2 - \text{mod}^2(n-1, 2)}{4}.
\end{aligned}$$

Converting it back to N (the number of amino acids), we finally arrive at

$$a_N = \frac{(N-4)^2 - \text{mod}^2(N-6, 2)}{4} \quad (N \geq 4).$$